

Range constructor for `std::string_view`

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1 Abstract

This paper proposes that `string_view` be constructible from any contiguous range of characters. The idea was extracted from P1206.

2 Tony tables

Before	After
<pre>void foo(string_view); vector<char8_t> vec = get_some_unicode(); foo(string_view{vec.data(), vec.size()});</pre>	<pre>void foo(string_view); vector<char8_t> vec = get_some_unicode(); foo(vec);</pre>

3 Motivation

While P1206 gives a general motivation for range constructors, it's especially important for `string_view` because there exist in a lot of codebases string types that would benefit from being convertible to `string_view`. For example, `llvm::StringRef`, `QByteArray`, `fbstring`, `boost::container::string` ...

Manipulating the content of a vector as a string is also useful.

Finally, this makes contiguous views operating on characters easier to use with `string_view`.

4 Design considerations

- instantiations of `basic_string` are specifically excluded because `std::basic_string` already provides a conversion operator and more importantly, strings with different `char_traits` should not be implicitly convertible
- Because `basic_string_view` doesn't mutate the underlying data, there is no reason to accept a range by something other than const lvalue reference.

- The construction is implicit because it is cheap and a contiguous range of character is the same platonic thing as a `string_view`.

5 Arrays and null terminated strings

During review by LWG, it was noticed that the proposed change introduces this arguably surprising behavior:

```
char const t[] = "text";
std::string_view s(t); // s.size() == 4;

std::span<char> tv(t);
std::string_view s(tv); // s.size() == 5;
```

This is not an ambiguity of the overload set but rather a consequences of both null-terminated terminated strings and array of characters being both sequence of characters with array of characters implicitly convertible to pointers.

To be consistent with C++17 and not introduce a behavior change, we make sure arrays of characters decay to `const charT*`. We think this proposed design is consistent with existing practices of having to be explicit about the size in the presence of embedded nulls as well as the general behavior of C functions, and does not introduce a new problem - how unfortunate that problem might be. It is also worth noting that while embedded nulls have a lot of known usages they are not the common case.

Finding a better solution to that problem is not possible at the level of this proposal and would require major breaking language changes.

6 Proposed wording

Change in `[string.view]` 20.4.2:

```
template<class charT, class traits = char_traits<charT>>
class basic_string_view {
public:
    [...]

    // construction and assignment
    constexpr basic_string_view() noexcept;
    constexpr basic_string_view(const basic_string_view&) noexcept = default;
    constexpr basic_string_view& operator=(const basic_string_view&) noexcept = default;
    constexpr basic_string_view(const charT* str);
    constexpr basic_string_view(const charT* str, size_type len);
```

```

template <class R>
constexpr basic_string_view(const R& r);

template <class It, class End>
constexpr basic_string_view(It begin, End end);

[...]
};
template<class R>
basic_string_view(const R&)
-> basic_string_view<ranges::range_value_t<R>>;
template<class It, class End>
basic_string_view(It, End) -> basic_string_view<remove_reference_t<iter_reference_t<It>>>;

```

Change in [string.view.cons] 20.4.2.1:

Add after 7

```

template <class R>
constexpr basic_string_view(const R& r);

```

Constraints:

- const R satisfies ranges::ContiguousRange,
- const R satisfies ranges::SizedRange,
- is_same_v<ranges::range_value_t<const R>, charT> is true,
- is_convertible_v<const R&, const charT*> is false,
- If the qualified-id R::traits_type is valid and denotes a type, is_same_v<R::traits_type, traits> is true.

Expects:

- const R models ranges::ContiguousRange,
- const R models ranges::SizedRange.

Effects: Initializes data_ with ranges::data(r) and size_ with ranges::size(r).

Throws: What and when ranges::data(r) and ranges::size(r) throw.

```

template <class It, class End>
constexpr basic_string_view(It first, End last);

```

Constraints:

- It satisfies ContiguousIterator,
- End satisfies SizedSentinel<It>,
- is_same_v<iter_value_t<It>, charT> is true,
- is_convertible_v<End, size_type> is false.

Expects:

- `[first, last)` is a valid range,
- It models `ContiguousIterator`,
- End models `SizedSentinel<It>`.

Effects: Initializes

- `data_` with `to_address(first)`,
- `size_` with `last - first`.

Add a new section `[string.view.deduction]` to describe the following deduction guides:

```
template <class It, class End>
basic_string_view(It, End) -> basic_string_view<remove_reference_t<iter_reference_t<It>>>>;
```

Constraints:

- It satisfies `ContiguousIterator`,
- End satisfies `SizedSentinel<It>`.

```
template<class R>
basic_string_view(const R&)
-> basic_string_view<ranges::range_value_t<R>>>;
```

Constraints: `const R` satisfies `ranges::ContiguousRange`.